STRIVE Research Report

Problem Motivation

STRIVE addresses the gap between fastest-route navigation and safety-aware routing. The system estimates road-segment crash risk from historical incidents, road attributes, time, and weather, then exposes those scores through API endpoints and a route optimizer.

The practical goal is not to replace travel-time routing, but to give users and downstream applications a tunable safety weight so they can trade a small amount of travel time for lower predicted crash exposure.

Dataset Description

The current repository contains chronological train, validation, and test parquet splits produced by the M1/M2 pipeline. Split sizes are: Train: 162,860 rows, 50.1% positive rate; Val: 34,898 rows, 48.6% positive rate; Test: 34,900 rows, 50.9% positive rate.

Processed features are built from FARS-derived incident labels, OSM road attributes, weather defaults/live weather fields, and segment-level historical accident rates. The held-out test split remains separate from tuning and final training decisions.

Feature Engineering

The model consumes the shared FEATURE_NAMES schema from app/ml/features.py: hour_of_day, day_of_week, month, night_indicator, road_class, speed_limit_kmh, precipitation_mm, visibility_km, wind_speed_ms, temperature_c, rain_on_congestion, historical_accident_rate.

This shared feature contract keeps M2 training, M3 risk inference, and M4 route edge scoring aligned so the same transformations are used in batch training and live API scoring.

Model And Evaluation

Modelling Decisions

XGBoost was selected because it handles non-linear feature interactions, mixed feature scales, and class imbalance well while remaining fast enough for API-side batch scoring of road graph edges.

The end-to-end training script supports Optuna time-series cross-validation or reuse of models/best_params.json. Current best parameters include max_depth=3, learning_rate=0.1541, n_estimators=455, subsample=0.968, colsample_bytree=0.553, and min_child_weight=7.

Evaluation Results

Current held-out metrics are: AUROC 0.6942, AUPRC 0.7233, F1 0.6953, ECE 0.0058. The optimal classification threshold recorded for binary evaluation is 0.3480.

AUPRC, F1, and calibration meet the stated M2 targets in the current evaluation artifact. AUROC is below the target, so model discrimination remains a known limitation for future data/model work.

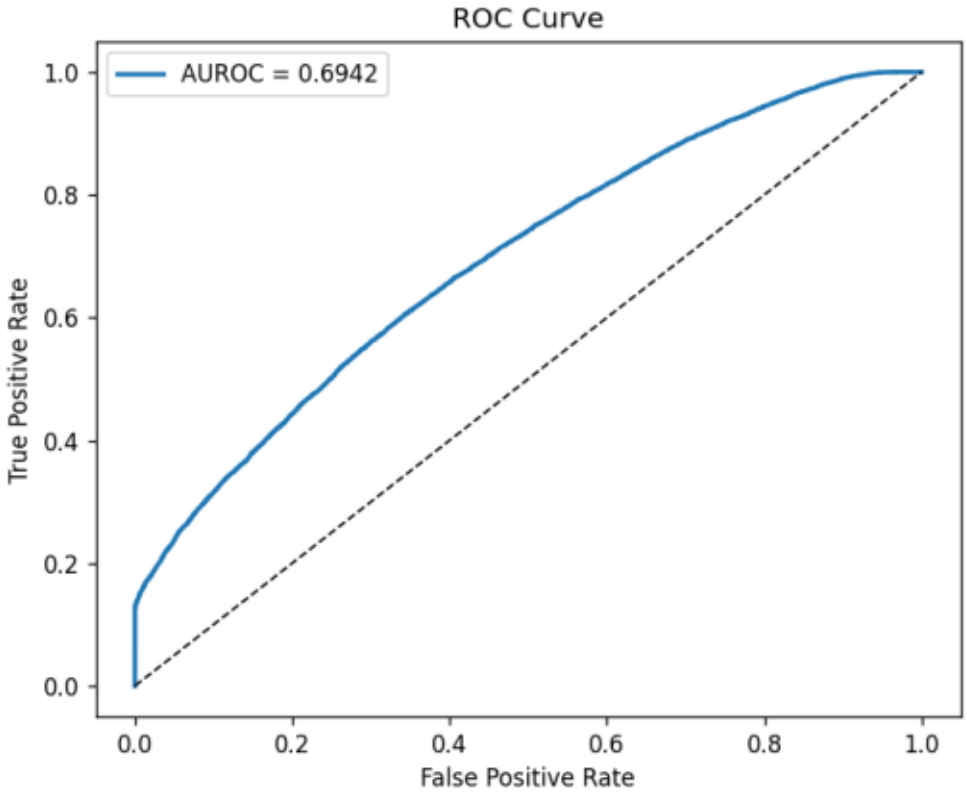
SHAP Analysis

Global SHAP importance highlights these top factors: 1. hour_of_day (0.9258); 2. night_indicator (0.9245); 3. month (0.1616); 4. day_of_week (0.1060); 5. speed_limit_kmh (0.0912).

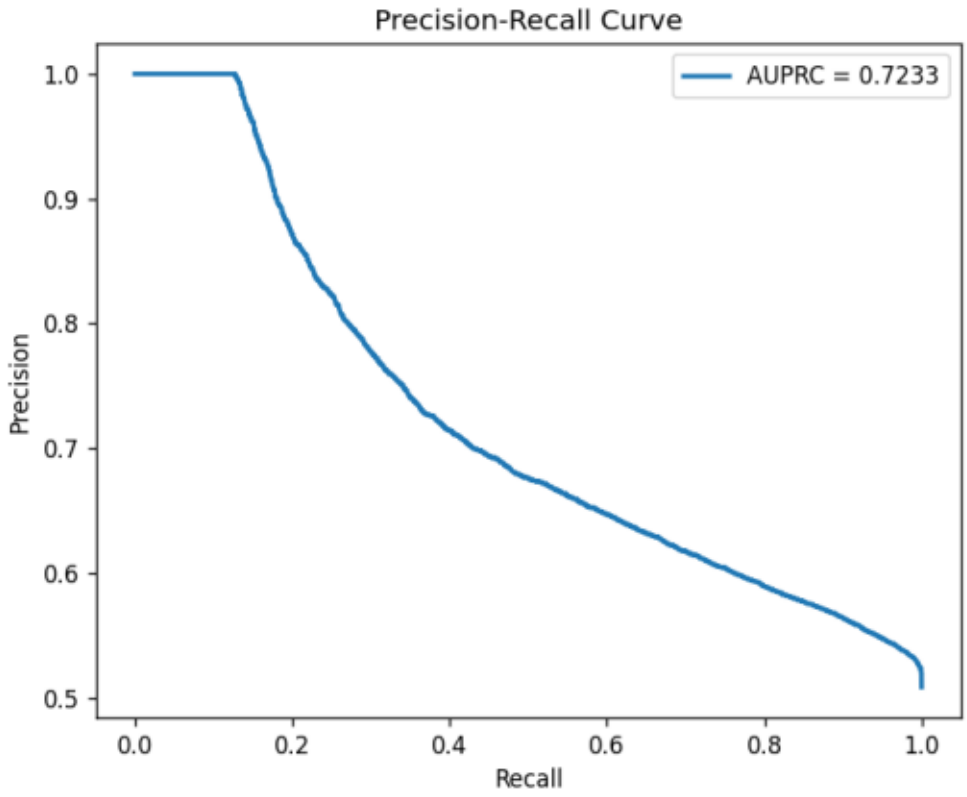
The current SHAP report validates precipitation for wet-night cases, but historical_accident_rate and night_indicator checks are not fully passing. This should guide future feature-quality and label-quality review rather than being hidden.

Evaluation Figures

roc_curve.png

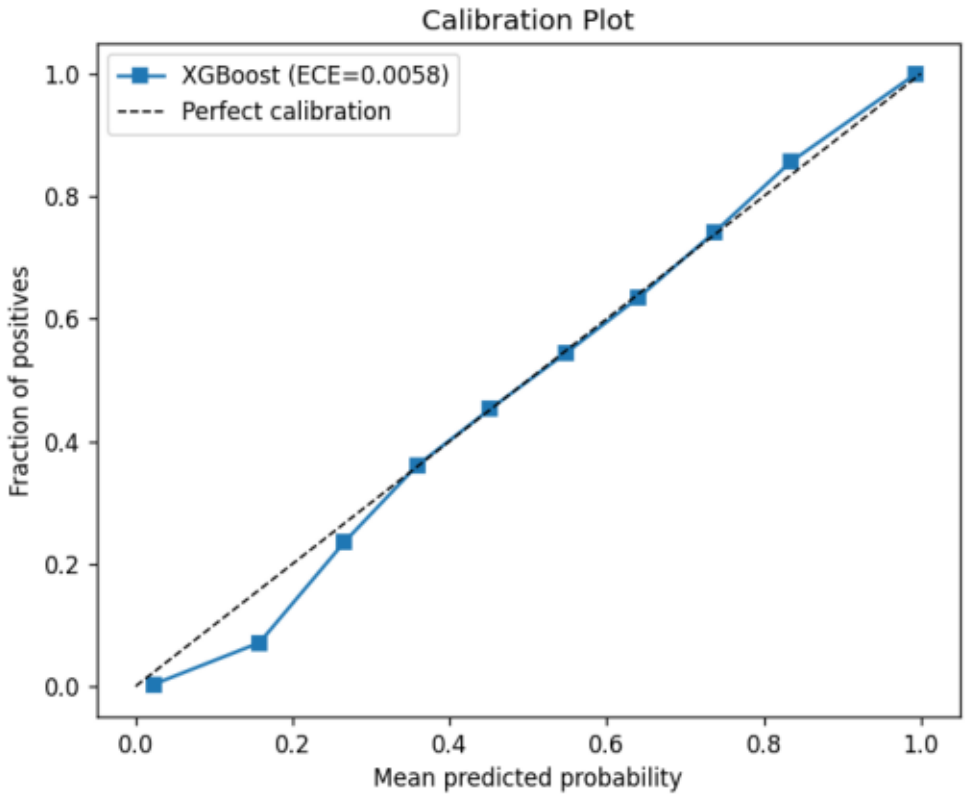


pr_curve.png

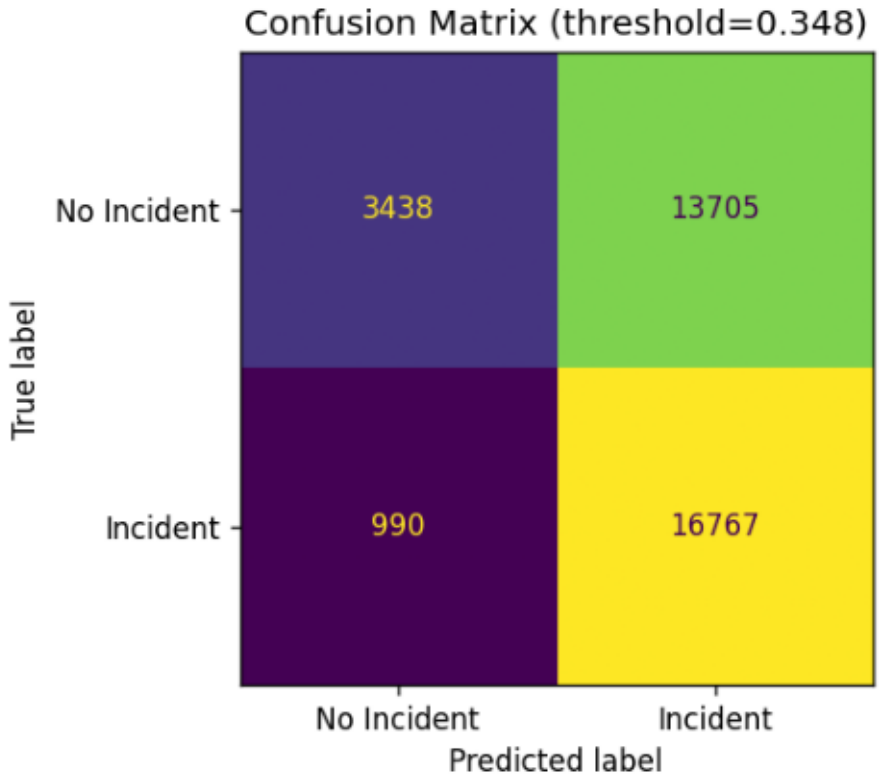


Calibration And Error View

calibration.png

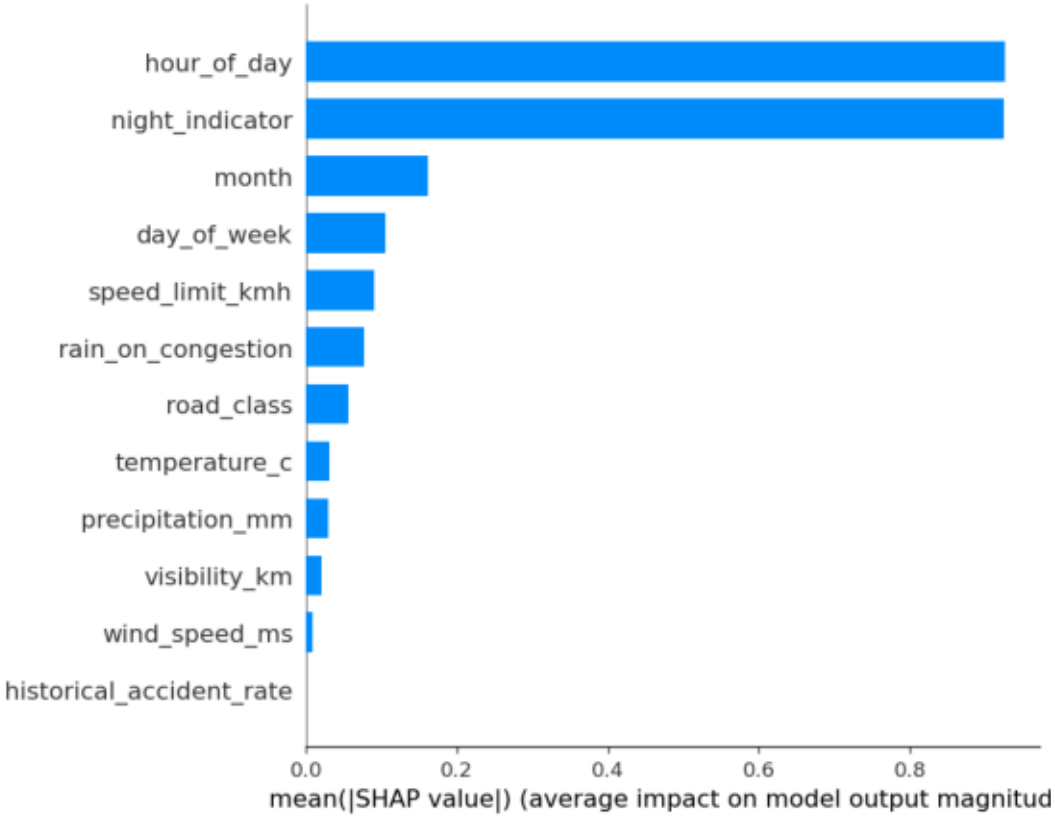


confusion_matrix.png

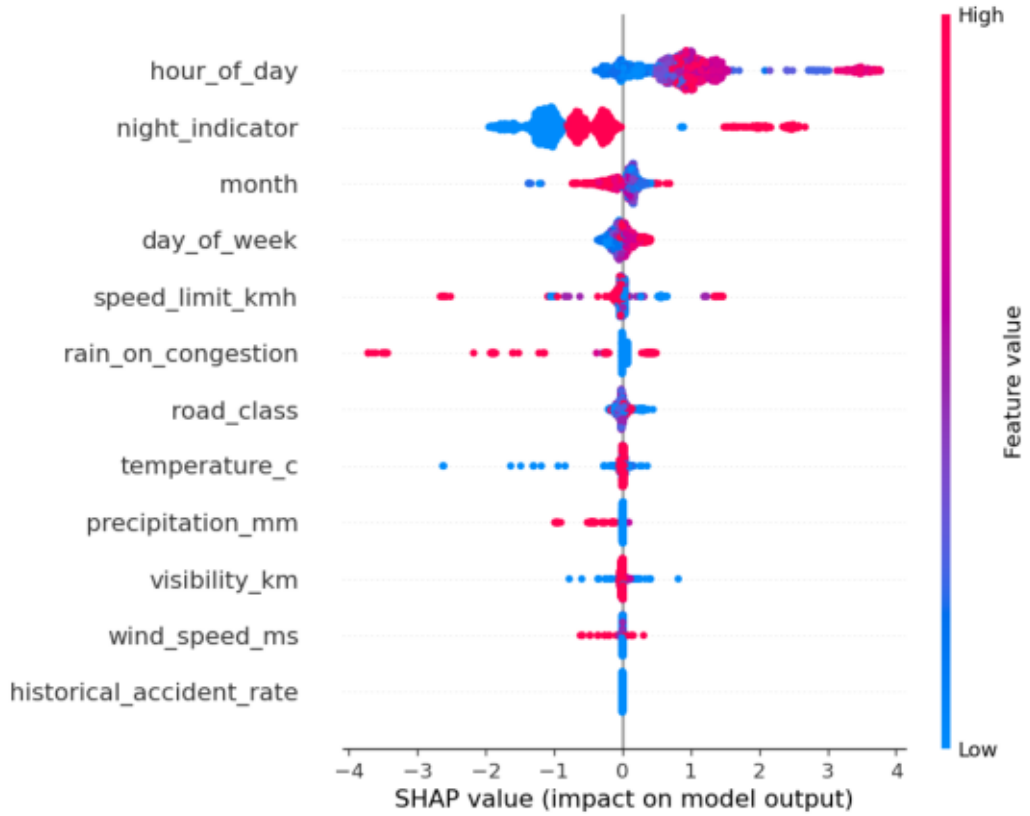


SHAP Figures

shap_summary.png



shap_beeswarm.png



Routing Quality And Next Steps

Routing Quality

M4 implements risk-weighted A* with cost = alpha * normalized_risk + (1 - alpha) * normalized_travel_time. The endpoint also computes a fastest-route baseline with alpha=0 and reports average risk reduction.

A controlled routing validation graph demonstrates the expected tradeoff: alpha=0 selects the shortest high-risk path, while alpha=1 selects the longer low-risk path. In that case, average segment risk drops from 95 to 5, a 94.7% reduction. The API smoke check also validates route generation on the local OSM graph.

Limitations

The current model artifacts are usable for API integration, but AUROC and two SHAP domain checks indicate that data enrichment and calibration should continue before treating scores as production-grade safety estimates.

Docker Compose files are present for API plus PostgreSQL, but Docker is not installed in this local environment, so container build verification must be run on a machine with Docker available.

Future Work

Seed PostgreSQL with full road segment metadata, cache scored edges by weather/time bucket, add route-level integration tests, and evaluate safe-vs-fastest routes on representative city trips.

Improve historical accident-rate coverage and collect richer negative samples so the model learns location risk beyond time-of-day proxies.